Meshaal Mouawad
Mississippi State University, Electrical and Computer Engineering Department,
This is a summary report of my research assignments I was given by my advisor during my PhD program

SUMMARY REPORT: FOR RESEARCH ASSIGNMENTS OF MY PHD

WEEK 10

Summary:

I. TSP Algorithms and DWA Method

Assignment:
1. TSP (Travelling salesman problem)
2. Study Miller Tucker Zelmin.
3. Local navigator using DWA (Dynamic Window Approach) method to walk a mobile robot from starting point to goal along the edges of built GVD.
4. Run the TSP MATLAB code

Completed: No
Submitted:
Problems: TSP matlab code is not working
Explain: NA
Procedure: The goal is to understand VD and TSP. TSP will be performed by Miller Tucker Zelmin (MTZ), then DWA will connect the points for mapping (the next assignment.

For the TSP code the adjacent matrix must be obtained from the VD which is in the MATLAB code file of VD. In the Matlab code of VD → Output → edge_matrix.

I run the code using both Spare version and non sparse and the TSP is not running. Tingjun, also tried it with me and we could not figure it out.

II. Read Elakiya's thesis

Assignment:
1. Read Elakiya's thesis and try to fully understand it. **Now please read parts from Chapter 5, Page 88** to end. Then read from beginning to end.
2. Please read and try to understand **Elakiya's defense presentation slides**.
3. Read and try to understand **Elakiya's conference paper.**

Completed: Yes
Submitted: NA
Problems: No
Explain: NA
Procedure: The goal is to understand VD and TSP fully.

III. ASSIGNMENT #2 VORONOI DIAGRAM

Assignment: Please read textbooks ONLY for chapter of "Path Planning and Navigation".
- "Introduction to Autonomous Mobile Robots", Roland Siegwart, Illah Reza Nourbakhsh and Davide Scaramuzza, Second Edition, MIT Press. (**Chapter 6 Planning and Navigation**).
- "Robotics, Vision and Control: Fundamental Algorithms in MATLAB®", Peter I. Corke, Springer 2017, ISBN: 978-3-319-54412-0.(**Only read path planning and navigation**).

Completed: No
Submitted: NA
Problems:
Procedure:

This Report was submitted in 2019 for Mississippi State University, Electrical and Computer Engineering Department during Fall 2019 foe the course of Introduction to Radars Engineering instructed by Dr. Ball John
Meshaal Mouawad, Mississippi State University, Starkville, MS 39759 USA (e-mail: mm4922@msstate.edu).

Appendix